Impact of Venus–Mercury orbital resonance on Earth's climate

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The Venus–Mercury orbital relationship could have an indirect influence on Earth's long-term climate through gravitational perturbations of the Solar System. However, there is no established evidence that a Venus–Mercury orbital resonance directly drives Earth's climate or causes major ice-age cycles.

The interesting scientific question is whether the gravitational interaction between Mercury and Venus can modify Earth's orbital parameters and thereby influence the distribution of solar radiation reaching Earth.

This involves three interconnected mechanisms:

1. Orbital resonance: gravitational interactions between Mercury and Venus.
2. Secular perturbations: slow changes in planetary orbital eccentricities and inclinations.
3. Milankovitch cycles: changes in Earth's eccentricity, axial tilt and precession, which influence long-term climate.

1. Venus–Mercury orbital relationship

Mercury Globe: 0°N, 270°E - NASA Science

Mercury

Orbital period: 87.97 days

Venus - Computer Simulated Global View Centered at 90 Degrees East Longitude | NASA Jet Propulsion Laboratory (JPL)

Venus

Orbital period: 224.70 days

Their orbital-period ratio is:

$$\frac{P_V}{P_M}=\frac{224.70}{87.97} \approx 2.554$$

This is relatively close to a 5:2 ratio (2.5), but it is not an exact 5:2 mean-motion resonance.

The distinction is important: proximity to a simple integer ratio does not, by itself, establish a dynamically significant resonance.

2. The important mechanism: secular resonance

The more scientifically significant connection is not the approximate 5:2 orbital-period ratio. It is a phenomenon called secular resonance, involving the slow rotation of planetary orbital ellipses and orbital planes.

Astronomers describe these slow frequencies using:

- g_1: Mercury's dominant apsidal precession frequency.
- g_2: Venus's dominant apsidal precession frequency.
- s_1,s_2: associated nodal precession frequencies.

A particularly important resonance is:

$$\boxed{\sigma_{12}=(g_1-g_2)+(s_1-s_2)\approx 0}$$

This resonance is dominated by Mercury and Venus.

Its importance was demonstrated in research by Richard E. Zeebe and Margriet L. Lantink (2024).

They found that this resonance can disrupt the familiar long eccentricity cycle of Earth, traditionally associated with a period of approximately 405,000 years.

This is a significant discovery for understanding Earth's geological climate history.

arXiv+1

3. How this affects Earth's climate

The causal chain is:

Mercury–Venus gravitational interactions

Long-term changes in orbital precession frequencies

Secular resonance σ_{12}

Changes in the structure of planetary orbital dynamics

Earth's orbital eccentricity spectrum changes

Amplitude and periodicity of eccentricity variations change

Milankovitch forcing changes

Changes in seasonal and latitudinal solar insolation

Long-term climate consequences

Potential changes in glaciation, ice-sheet dynamics and sedimentation

There are three important climate variables involved:

Orbital parameter	Climate consequence
Eccentricity	Modulates Earth's orbital shape and seasonal insolation
Obliquity	Changes the strength of seasonal contrasts, especially at high latitudes
Precession	Changes the timing of seasons relative to Earth's position in its orbit

These are the foundations of Milankovitch climate forcing.

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4. The 405,000-year climate cycle

One particularly interesting result concerns the approximately 405-kyr eccentricity cycle.

This cycle is conventionally associated with:

\[g_2-g_5 \]

where \(g_2\) is dominated by Venus and \(g_5\) by Jupiter.

The Mercury–Venus secular resonance can indirectly disrupt this cycle.

Zeebe and Lantink's numerical integrations found:

- The 405-kyr cycle can become extremely weak or disappear.
- Its periodicity and amplitude are not necessarily stable over geological time.
- Episodes of the relevant secular resonance occurred in approximately 40% of their long-term numerical solutions.
- Earth's climate-forcing spectrum can become substantially different during these episodes.

This is particularly important because geologists have used the 405-kyr cycle as a geological dating reference or astronomical metronome.

arXiv+1

5. Does this explain ice ages?

Potentially, yes, but indirectly.

Earth's glacial and interglacial cycles are influenced by orbital forcing, particularly:

- Approximately 41,000-year obliquity cycles.
- Approximately 19,000–23,000-year precession cycles.
- Approximately 100,000-year eccentricity-related climate cycles.

The Mercury–Venus secular resonance can modify the long-term astronomical forcing underlying these cycles.

However, it does not mean that Mercury and Venus directly trigger ice ages. The climate response also depends on carbon dioxide, ocean circulation, ice-sheet dynamics, vegetation and other feedback mechanisms.

6. Implications over different timescales

Timescale	Potential significance
Decades to centuries	No established significant climate effect
10,000–100,000 years	Indirect modification of orbital climate forcing
100,000–1 million years	Changes in eccentricity cycles and glacial pacing
Millions to billions of years	Important changes in the stability and structure of astronomical climate forcing

One particularly interesting implication is that the astronomical forcing responsible for Earth's ancient climate may have been substantially different from today's forcing, even without major changes in planetary orbits.

7. Research papers worth examining

These are particularly relevant to your question:

1. Zeebe & Lantink (2024) – A secular solar system resonance that disrupts the dominant cycle in Earth's orbital eccentricity (g2-g5): Implications for astrochronology.

Read the research paper (arXiv).

2. Zeebe & Lantink – Milanković Forcing in Deep Time.

Read the research paper (Paleoceanography and Paleoclimatology).

3. Mapping Solar System Chaos with the Geological Orrery (2019).

Read the research paper (PNAS).

8. An interesting computational research direction

There is an opportunity to investigate this problem using numerical celestial mechanics.

I would propose a three-stage model:

1. N-body planetary simulation: Integrate Mercury, Venus, Earth, Mars, Jupiter and Saturn using Newtonian gravitational equations, with relativistic corrections particularly important for Mercury.
2. Frequency analysis: Extract \(g_1,g_2,g_5,s_1,s_2\), identify episodes of \(\sigma_{12}\) resonance, and measure changes in Earth's eccentricity spectrum.
3. Paleoclimate correlation: Compare calculated eccentricity, obliquity and precession with geological climate proxies, including marine sediment records and oxygen-isotope records.

This would help quantify how much of Earth's long-term climate variability is attributable to changes in Mercury–Venus secular dynamics.

My central conclusion: The Venus–Mercury gravitational relationship is potentially important to Earth's climate history, not because it significantly changes the amount of sunlight Earth receives directly, but because it can modify the orbital dynamics that govern Milankovitch climate cycles. The 2024 research provides particularly compelling evidence for this connection over geological timescales.